JASON YAO

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EDUCATION

New York University
College of Arts & Science
BA in Computer Science
Grad. May 2017

SKILLS

Programming

Languages
Kotlin, Java, Python, C++, TypeScript, C, Bash

AI/ML
PyTorch, TensorFlow, Diffusers

Web Frameworks
Spring/SpringBoot, Spring Reactor/WebFlux, Django, NodeJS, Vue 3, OpenAPI, Protobuf

Databases & Processing
PostgreSQL, MongoDB, Spanner, Redis, Hibernate, JPA, R2DBC, Kafka, AWS SQS

Observability
Prometheus, Grafana, LightStep, OpenTelemetry

Tools
git, vim, Kubernetes, Docker, Github Actions, LaTeX, Consul

General

Miscellaneous
I know how to learn, use Google, sing badly in the shower, and read SO



11.08.2026

EXPERIENCE

Google — New York, NY (5 years, 2 months)

Staff Software Engineer — Nov 2026 – Present

- **Tech lead for a 13-engineer team** and cross-org working group lead driving architectural roadmap, trace governance, and reliability across Google Search's debugging and distributed tracing infrastructure
- Authored, led, and executed an 8-workstream trace governance and automated culprit-finding architecture, **reducing volume regression diagnosis and rollback from mutli-week incidents to minutes**

Senior Software Engineer — Nov 2023 – Nov 2026

- Spearheaded cross-org distributed tracing stabilization across Search, Evaluations, and Core, **reducing Search trace emissions by 33.1%, cutting p50 latency by 25.7%, and raising eval trace reliability from 15% to 90%**
- Productionized an automated protobuf fragmentation and compression pipeline, **reducing emitted debug data volume by 92%**; architected adaptive eval trace sampling for a **64% trace size reduction**
- Architected and migrated Search's core debug logging SDK (**11k+ LoC across 100+ PRs**) to introduce team attribution, load shedding, and child context primitives with **zero downstream migration burden**
- Designed and led targeted debug data caching, **decreasing latency by 5x, and reducing error rates from 11% to ~0.1%**
- Mentored team members, conducted 80+ interviews, and dozens of Hiring Committee reviews

Software Engineer — Sep 2021 – Nov 2023

- Proposed, designed, and led team of 5 SWEs to overhaul Search debugging performance and platform architecture, delivering a **99.8% error rate reduction, ~70% latency improvement,** and **25+ SWE-years saved annually**

Squarespace — New York, NY (3 years, 6 months)

Software Engineer III (Domains) — Mar 2018 – Sep 2021

- Designed and built registrar TLS, EPP, and product features that **lowered domain registrations from ~30s to ~100ms (99.6% latency reduction)**
- Designed and built robust backend services and systems for Squarespace Emails that **lowered error rates from ~40% to ~2%,** and **decreased customer support volume from ~60% to <1%**
- Proposed, designed, and built the backend systems for a generic fraud detection system, **increasing the number of correctly-identified fraudulently purchased domains by over 3.5x baseline.** Converted fraud detection model at Squarespace from *reactive to customer write-ins* to *proactively identifying fraudulently purchased domains*